Valuation and Satiation: A Two-Dimensional Theory of Nonlinear Pricing

Bing Liu*

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Abstract

Why do firms use such varied non-linear pricing? I show the full range of patterns arises from the correlation between consumer valuation and satiation, including increasing or zero marginal prices—patterns that canonical models cannot explain. The negative correlation case violates the single-crossing assumption. I characterize the optimal mechanism as a multi-part tariff that exhibits several non-standard outcomes, including non-monotonic allocations, monopoly overprovision, and full consumer surplus extraction.

*Liu: Department of Economics, Stanford University. (bingliu@stanford.edu). I thank Paul Milgrom, Ilya Segal, Ravi Jagadeesan and Alvin Roth for their invaluable mentorship and advice. I also thank Roberto Corrao, Laura Doval, Pablo Schenone, Vijay Krishna, Modibo Camara, Kristof Madarasz, Rene Kirkegaard, Yi Chen, Martino Banchio, Andrés Perlroth, Spencer Pantoja, Jason Weitze, Mitchell Watt, Alexander Haberman, Klajdi Hoxha, Danielle Handel, Dorsa Majdi, Filip Tokarski, Eric Tang, Chanjoo Lee, Alex Tordjman, Joey Feffer, as well as audiences at the Canadian Economic Theory Conference and the Stanford Market Design Seminar for their especially helpful comments and suggestions.

Non-linear pricing is ubiquitous in practice. It manifests in everyday transactions, from the varying prices of different-sized soft drinks and the tiered rates for utility consumption (e.g., water, data) to the variable fees for shipping packages. Understanding non-linear pricing is therefore core to analyzing how firms with market power maximize profits. For policymakers, explaining why these pricing patterns emerge is also central to diagnosing market inefficiencies and shaping effective regulation and welfare analysis.

However, existing theory does not yet provide a unified account of the variety of non-linear pricing schedules observed in practice. In particular, two commonly observed patterns sit uneasily with the canonical monopoly non-linear pricing framework. The first is the observation of zero marginal prices. In standard models (e.g., Mussa and Rosen (1978), Maskin and Riley (1984), Wilson (1993)), optimal tariffs are typically smooth and feature strictly positive marginal prices almost everywhere. Yet zero marginal prices are increasingly common: consumers pay nothing at the margin for an additional hour of streaming under a Netflix subscription, an additional song on Spotify, an additional ride with a seasonal ski pass, or an additional gym visit under a flat-fee membership. Existing explanations often rely on departures from the benchmark profit-maximizing screening model, such as biased or inconsistent beliefs about future demand (Grubb, 2009) or a non-profit component in the seller's objective (Corrao et al., 2023). A second missing piece is the prevalence of quantity premium—marginal prices that increase with quantity. While Maskin and Riley (1984) shows that quantity discounts are optimal almost everywhere under mild conditions, tariffs with quantity premium are widespread, for example in utility pricing: in Redwood City, California, residential water is priced in tiers with higher per-unit rates at higher total consumption,[1] and PG&E's tiered electricity plan similarly charges a higher per-kWh rate above a baseline allocation. Analogous quantity premium also appears in other markets—for instance in tiered plans in MoviePass[2] and DHL shipping.

This paper develops a unified framework of monopoly pricing that explains the full set of pricing patterns described above – both zero and positive marginal price, as well as marginal prices that increase with quantity (quantity premium) and decrease with quantity (quantity discount). To derive an optimal pricing strategy, a firm must understand how consumer utility changes with quantity. Two aspects of the utility function are critical: (i) marginal utility for each additional unit, which is crucial for the local incentive compatibility constraint; and (ii) how this marginal utility varies, and usually diminishes, with quantity, which is crucial for the local incentive compatibility constraint. This paper

[1] https://www.redwoodcity.org/departments/public-works/water/rates/current-rates
[2] https://www.moviepass.com/plans?cookie_consent=false

reduces these two pieces of information into two dimensions of consumer type: a per-unit valuation v and a finite satiation quantity d. The per-unit valuation captures the consumer's initial marginal utility while the finite satiation quantity determines how fast the marginal utility diminishes with quantity. In practice, the correlation between these two dimensions can be positive or negative. For example, the correlation tends to be positive in the electricity market: consumers who value each kWh more typically own more appliances and have a higher satiation quantity. Conversely, the correlation is often negative for streaming services, as users with a high willingness-to-pay may also have a high opportunity cost of time, thus exhibiting a lower satiation quantity. This paper shows that a standard profit-maximizing framework that permits a broad range of correlations between valuation and satiation can rationalize the full set of pricing patterns described above. In particular, quantity premiums and discounts arise when the two dimensions are positively and negatively affiliated, respectively. Furthermore, zero marginal price, which is a form of quantity discount, results from a particular 'weak'[3] negative correlation between the two dimensions.

Modeling consumer preferences along these two dimensions offers substantial analytical tractability. In Lemma 3, we establish a monotone comparative statics result: under any pricing schedule, a consumer with both a higher per-unit value and a higher satiation quantity weakly prefers a larger bundle than a consumer with both a lower per-unit value and a lower satiation quantity. This monotonicity in consumer behavior allows us to characterize the optimal cutoff type—the lowest per-unit value that purchases each quantity q. We then construct a pricing schedule that implements these optimal cutoffs. Our affiliation condition is defined using the conditional distribution of per-unit values given that the satiation quantity is at least d, which aligns naturally with the monotone property of consumer behavior established in Lemma 3. The sign of this affiliation condition proves sufficient to determine the direction of quantity-based pricing.

Quantity premium are optimal when the two dimensions are positively affiliated. The intuition is twofold. First, consider the extreme case of perfect positive affiliation, where satiation quantity is a non-decreasing function of per-unit value. In this case, consumers with higher satiation quantities also have higher willingness to pay for each unit. The optimal allocation assigns larger bundles to these consumers, who under positive affiliation are precisely those with higher per-unit values. An increasing marginal price serves three purposes: it screens out consumers with lower per-unit values, directing them toward smaller bundles; it preserves incentive compatibility for high-value consumers choosing

[3]Loosely speaking, consumers with lower satiation quantity still have higher maximum willingness to pay (satiation quantity $\times$ per-unit value).

larger bundles; and it extracts additional revenue. Second, quantity premium improve the distribution of per-unit values for each quantity bundle when the two dimensions exhibit positive (but imperfect) affiliation. Under quantity premium, some consumers optimally purchase quantities below their satiation levels. This pricing structure thus serves a dual role: it enables the seller to charge higher marginal prices for larger bundles while simultaneously shifting the per-unit value distribution favorably at each quantity level. In this sense, rationing on the intensive margin is profitable for the seller. Positive affiliation between valuation and satiation helps explain quantity premium in settings such as utility pricing and subscription services like MoviePass: consumers with higher per-unit values—those who value the convenience of electric appliances more or appreciate movies in theaters more—tend to have higher satiation quantities.

Quantity discounts are optimal when the two dimensions are negatively affiliated. Under negative affiliation, a consumer is less likely to have high per-unit value conditional on having a high satiation quantity. Rationing on the intensive margin through quantity premiums is no longer profitable because there is insufficient probability mass of high per-unit value consumers with high satiation quantities. Instead, it becomes optimal for the seller to sell all participating consumers their full satiation quantity. Consider, for example, the case of perfect negative correlation. Consumers with higher marginal utility (per-unit value) have lower satiation quantities. The seller would prefer to sell larger quantities to high-satiation consumers, but these are precisely the consumers with lower per-unit values. Incentive compatibility then forces marginal prices to decline with quantity—quantity discounts are, in this sense, inevitable.

The natural next question is what the discount looks like—and what economic force shapes it. To characterize the shape of the tariff under perfect negative correlation, a useful object is the consumer's maximum surplus—the most the seller could ever extract from that consumer—defined as the product of per-unit value and satiation quantity. We say the two dimensions weakly (respectively, strongly) negatively correlated if maximum surplus increases (respectively, decreases) with per-unit value. With strong negative correlation, higher per-unit value implies lower maximum surplus. The optimal tariff—total payment as a function of quantity—coincides with the maximum-surplus schedule as a function of satiation quantity. This yields an *all-you-can-pay region*: full surplus extraction is feasible and each consumer purchases exactly her satiation quantity. No consumer buys less than satiation, because doing so would imply paying an average per-unit price above her value; and no consumer buys more, because that would require a payment exceeding her maximum surplus. Strong negative correlation therefore allows the seller to perfectly separate types. With weak negative correlation, by contrast, the seller can-

not (and optimally does not) fully separate types. The optimal tariff features an *all-you-can-eat* region in which consumers pay the same total price but consume their satiation quantities. In this region, incentive constraints bind: any tariff that is increasing in per-unit value would imply a higher total payment for a smaller quantity, violating incentive compatibility. Overall, the optimal pricing schedule is a hybrid of all-you-can-pay and all-you-can-eat. In both the all-you-can-pay and all-you-can-eat regions, high-value consumers end up consuming relatively low quantities.[4] Quantity discounts therefore use quantity as a screening device to cream-skim high-value consumers. The resulting tariff closely mirrors real-world pricing schedules, such as tiered data plans and passes for ski resorts and gyms. The all-you-can-eat component corresponds to the zero marginal prices of buffets and to flat-rate subscription models on platforms such as Netflix and Spotify. In this model, quantity discounts serve to maximize the surplus captured by the seller: smaller bundles carry higher per-unit prices to extract surplus from high-value, low-satiation consumers, while larger bundles feature higher total payments to extract surplus from consumers with lower per-unit values but higher maximum surplus. Zero marginal pricing arises under weak negative correlation, when quantity is no longer an effective screening instrument and screening instead operates primarily through the *total* payment. This also provides intuition for monopoly overprovision: by supplying a quantity above the socially efficient level, the monopolist raises revenue by more than it increases social surplus, because the extra quantity sharpens screening and allows greater surplus extraction from high-value types.

This framework sheds light on why the standard non-linear pricing models do not predict quantity premium or zero marginal price. The issue lies in their implicit restrictions on how consumer utility vary with quantity—especially how their valuation and satiation correlate. The canonical one-dimensional screening model relies on the single-crossing property that, intuitively speaking, implies that higher-valuation consumers also have higher satiation. The conditions in Maskin and Riley (1984) under which quantity discounts are optimal further impose curvature conditions on the utility functions. It requires[5] the existence of consumption levels $q_1 < q_2$ such that the marginal utility of the higher valuation consumers at q_2 is lower than the marginal utility of the lower valuation consumers at q_1. This paper's two-dimensional approach complements the standard literature precisely where the latter is most restrictive. On the negative-correlation side, it characterizes optimal pricing when the single-crossing property fails. Proposition 8

[4]This differs from the rationale for quantity discounts in Maskin and Riley (1984), where high-value consumers end up consuming high quantities.

[5]This is a necessary but not a sufficient condition for the quantity discount result in Maskin and Riley (1984).

further shows that the single-crossing assumption is the key force behind the absence of zero marginal pricing. On the positive-correlation side, it provides a simple environment—linear utility with positive affiliation—in which the conditions in Maskin and Riley (1984) are violated in a natural way, thereby rationalizing quantity premium.

Under perfect negative correlation, the single-crossing condition fails. As a result, the standard envelope-theorem approach is no longer applicable. This paper develops a method to solve the problem in this setting. It involves two steps. First, approximate the maximum-demand function by simple (step) functions—finite linear combinations of indicator functions—and, by taking the limit as the approximations refine, characterize a key feature of the optimal mechanism: quantity discount. Conditional on the quantity-discount schedule, we apply the standard principle of maximizing the seller's extraction of consumer surplus subject to incentive-compatibility and individual rationality constraints. In this particular setting, it implies designing a payment function that is non-increasing in the consumer's per-unit value to ensure incentive compatibility, while capturing as much surplus as possible. The resulting payment function is a hybrid of *all-you-can-pay* and *all-you-can-eat*. In contrast with the standard mechanism-design approach, where one typically derives the optimal allocation first and then recovers payments via the envelope theorem, in our model the order is reversed: we first pin down the optimal payment schedule and then derive the allocation that implements those payments. Crucially, the optimal allocation need not be monotone in a consumer's per-unit valuation; it can be non-monotonic. An important implication is that in a monopoly market where consumers' satiation quantity and per-unit valuation are negatively correlated, intermediate types may be excluded from the market even though lower- and higher-valuation consumers are served.

This paper builds upon the extensive literature on non-linear pricing under the single-crossing assumption, including seminal works such as Mussa and Rosen (1978), Maskin and Riley (1984), and Goldman et al. (1984). Comprehensive reviews of this literature can be found in Wilson (1993) and Armstrong (2016).

Our result on the optimality of zero marginal pricing complements existing explanations by Corrao et al. (2023) and Grubb (2009). In their models, zero marginal price can be optimal because either the seller derives intrinsic value from user usage (Corrao et al. (2023)), or because consumers hold biased beliefs and underestimate the variance of their future demand (Grubb (2009)). In contrast, our model generates this outcome within the standard framework, requiring neither alternative seller motivations nor consumer misperception. TThe result on the optimality of zero marginal pricing (all-you-can-eat tariffs) contributes to the literature on optimal bundling, which traces back to Stigler (1963),

Adams and Yellen (1976), and McAfee et al. (1989), and more recently includes Ghili (2023), Haghpanah and Hartline (2021), and Yang (2025). In our model, different quantity bundles can be reinterpreted as different levels of nested bundles; thus, the optimality of zero marginal pricing is equivalent to the optimality of pure grand bundling. The intuition underlying our result closely parallels that of Haghpanah and Hartline (2021): zero marginal pricing (pure bundling in their framework) is optimal when the consumer with higher per-unit value (higher value for small bundles) also has higher maximum surplus (higher value for the grand bundle).

On the technical side, our work contributes to body of research that explores violations of the single-crossing assumption , such as Araujo and Moreira (2010), Araujo et al. (2015), Schottmüller (2015), Chen et al. (2022), and Kwak (2022). With the exception of Kwak (2022), which identifies conditions under which non-single-crossing problems can be reduced to those satisfying single-crossing, these papers introduce alternative conditions, such as double-crossing (Chen et al. (2022)), inversely U-shaped decision functions (Araujo et al. (2015)), and monotone decision functions (Schottmüller (2015)), to ensure global incentive compatibility alongside local incentive compatibility. In contrast, our paper adopts a practical approach, acknowledging that consumer preferences in real-world markets often violate single-crossing. By incorporating a satiation quantity that negatively correlates with per-unit value—a crude approximation of real-world, non-single crossing preferences—our model predicts optimal non-linear pricing strategies that align with the prevalence of multi-part tariffs with unlimited options and subscription models. Furthermore, we show the optimality of these pricing schemes for consumer preferences that violate single-crossing in more general forms, beyond the satiation framework. Within the satiation framework, we show how the violation of single-crossing can alter standard economic intuitions.

The paper is organized as follows. Section 1 introduces the model. Section 2 characterize the optimal pricing. Section 3 discusses how our results extends beyond the satiation model. All proofs are in the appendix.

1 Model

There is a seller and a consumer. The seller can supply a homogeneous good at a constant marginal cost of $c \geq 0$. For most of the paper, we let $c = 0$ to focus on the effect of consumer's preferences.

The consumer's type and payoff. The consumer has a two-dimensional type, $\theta = (v, d)$, where $v \in [\underline{v}, \overline{v}]$ is the consumer's per-unit valuation of the good and $d \in [\underline{d}, \overline{d}]$ is the consumer's satiation quantity. For a given type (v, d), an allocation q and a payment T, the consumer's payoff is

$$v \min\{q, d\} - T.$$

So the consumer has a constant marginal utility of v for all units $q \leq d$ and a marginal utility of 0 for all units $q > d$. In other words, the consumer is fully satiated at quantity d and derives no more utility from any additional units. The consumer's type (v, d) is her private information and follows the joint distribution $F(\cdot, \cdot)$ with density $f(\cdot, \cdot)$.

The conditional virtual type and the conditional cut-off type. Two quantities related to the consumer's type distribution are important to the paper. We define them here. For a consumer of type (v, d), let $\phi(v; \geq d) := v - \frac{1-F(v;\geq d)}{f(v;\geq d)}$ denote her conditional virtual type. Throughout the paper, we assume for any d, $\phi(v; \geq d)$ is non-decreasing in v. This is analogous to the regularity condition in Myerson (1981). By Myerson (1981), a monopolist facing a consumer with a value distribution $F(v; \geq d)$ will optimally post a price equal to $\hat{v}(d)$ where $\hat{v}(d)$ is such that $\phi(\hat{v}(d); \geq d) = 0$.

Correlation between v and d. A key result of the paper provides sufficient conditions on the correlation between v and d under which quantity premiums or quantity discounts are optimal. To that end, we define two kinds of correlations between v and d: when v and d are positively affiliated and when v and d are negatively affiliated.[6]

Definition 1. *v and d are positively (respectively, negatively) affiliated if for all $d > d'$, $\frac{f(v;\geq d)}{f(v;\geq d')} \geq \frac{f(v';\geq d)}{f(v';\geq d')}$ (respectively, $\leq$) for all $v > v'$.*

The following result the relates the correlation of v and d and the conditional cut-off types is useful. We state it in Lemma 1.

Lemma 1. *If v and d are positively (respectively, negatively) affiliated, $\hat{v}(d)$ is non-decreasing (non-increasing) in d.*

Lemma 1 provides sufficient intuition for our main results about the quantity premium and quantity discount (Proposition 1 and Proposition 3). It demonstrates why positive and negative affiliation, as we define them, are the natural stochastic monotonicity

[6]Note that this definition of positive affiliation is different from the definition that v and d are affiliated à la Milgrom and Weber (1982). Neither our definition of positive affiliation implies affiliation à la Milgrom and Weber (1982) nor vice versa.

conditions that govern when these pricing patterns emerge. When the two dimensions are positively (respectively, negatively) affiliated, it is more (respectively, less) likely for a consumer with higher satiation quantity to have a higher per-unit value. Then the conditional cut-off type increases (respectively, decreases) in quantity.

The seller's problem. It is without loss to consider the seller offering a menu of quantity bundles $(q, t(q))_{q\in[0,\overline{d}]}$ to the consumer, with $t(q)$ being the per-unit price when the consumer purchases the q-unit bundle. The seller's objective is to maximize profit

$$\mathbb{E}[(t(q(\theta)) - c)\, q(\theta)]$$

subject to the consumer maximizing their own payoff, that is, for each θ

$$q(\theta) \in \arg\max_{q} \quad v \min\{d, q\} - t(q)q.$$

We use $\pi(t) := \mathbb{E}[(t(q(\theta)) - c)\, q(\theta)]$ to denote the seller's profit when he's using the average price function $t(\cdot)$.

Seller's problem under perfect correlation. Part of the paper also deals with the special case where the satiation quantity and per-unit value are perfectly correlated. In those cases, a consumer's type is $\theta = (v, d(v))$ for some monotone function $d(\cdot)$. This reduces the problem to a single-dimensional screening problem. We take the direct revelation approach to the problem. Let $q : [\underline{v}, \overline{v}] \to [0, \overline{d}]$ and $T : [\underline{v}, \overline{v}] \to \mathcal{R}$ be the allocation and total payment rule. The seller's problem is to choose (q, T) to maximize profit

$$\mathbb{E}[T(v) - cq(v)]$$

subject to (IC)

$$v \min\{d(v), q(v)\} - T(v) \geq v \min\{d(v), q(v')\} - T(v'), \quad \forall v, v' \in [\underline{v}, \overline{v}]$$

and (IR)

$$\theta \min\{d(v), q(v)\} - T(v) \geq 0, \quad \forall v \in [\underline{v}, \overline{v}]$$

Quantity premium/discount. We are interested in when the optimal pricing exhibits quantity premium (respectively, discount). Here we define the two types of pricing sched-

ules formally.

Definition 2. *The optimal pricing exhibits quantity premium (respectively, discount) if there exists a solution $t(\cdot)$ to the seller's problem such that $t(q)$ is non-decreasing (respectively, non-increasing) in q.*

To put simply, a quantity premium arises when larger purchases are charged a higher per-unit price, whereas a quantity discount occurs when larger quantities are priced lower on a per-unit basis.

It turns out that it is sometimes easier to characterize the solution in terms of the marginal price rather than the average price. Let $\tau(\cdot)$ be the marginal price associated with $t(\cdot)$. The following lemma presents an equivalence result.

Lemma 2. *The optimal pricing exhibits quantity premium (respectively, discount) if there exists a solution $t(\cdot)$ to the seller's problem such that its associated marginal price $\tau(q)$ is non-decreasing (respectively, non-increasing) in q.*

2 Characterizing the optimal pricing

We start with a straightforward observation about the monotonicity in the consumer's consumption choice given any pricing schedule. We state the observation in Lemma 3. The proof is straightforward from a monotone selection argument.

Lemma 3. *Given any pricing schedule $t(\cdot)$, if a consumer with type (v, d) purchases q units, then a consumer with type (v', d') with $v' \geq v$ and $d' \geq d$ purchases at least q units.*

Given Lemma 3, for each unit q, there exists a cut-off consumer type in the form of (v, q), such that all consumers with type (v', d) with $v' \geq v$ and $d \geq q$ purchases at least q units. Then the seller's problem is equivalent to choosing a cut-off type $v(q)$ for each of the q-th unit and a pricing schedule that can implements the optimal cut-off types $v(\cdot)$. From Myerson (1981), if a monopolist is simply solving for the optimal cut-off type to sell q units to a consumer with per-unit value distributed according to $F(v; \geq q)$, then the optimal cut-off type is $\hat{v}(q)$. The seller's actual problem imposes two further constraints on these cut-off types. First, the cut-off types $v(\cdot)$ must satisfy that for all (v, d) such that $v \geq v(q)$ and $d \geq d$, $v \geq v(q')$ for all $q' \leq q$. Second, there exists a pricing schedule that implements $v(\cdot)$. If there exists a pricing schedule that implements $v(q) = \hat{v}(q)$ and satisfies the first constraint, then that pricing schedule is optimal.

2.1 Positive affiliation and quantity premium

When v and d are positively affiliated, the pricing schedule with marginal price at q equal to the conditional cut-off type $\hat{v}(q)$ implements the cut-off types and is hence the optimal pricing schedule. Moreover, by Lemma 1, $\hat{v}(q)$ is non-decreasing in q, so the optimal marginal price increases in q and the optimal pricing exhibits quantity premium. We summarize this result in Proposition 1.

Proposition 1. *If v and d are positively affiliated, the optimal pricing exhibits quantity premium and $\tau(q) = \hat{v}(q)$.*

When the marginal price $\tau(\cdot)$ is non-decreasing in quantity, a consumer of type (v, d) optimally consume a quantity $q \leq d$ such that the marginal price at q is as close to v as possible from below. For each quantity q, the marginal type is (v, q) with $v = \tau(q)$. This is the exact reason why a price schedule with marginal price equal to the conditional cut-off type $\hat{v}(q)$ implements the cut-off type $\hat{v}(q)$.

Profitable rationing on the intensive margin. Under $\tau(\cdot)$, there is underprovision on both the extensive margin: some consumers consume zero; and the intensive margin: some participating consumers consume below their satiation quantities. Corollary 1 states this profitable rationing observation and Figure 1 provides an illustration.

Corollary 1 (Profitable rationing)**.** *When v and d are positively affiliated, under the optimal pricing, consumers of type $\theta = (v, d)$ where $v \in (\hat{v}(\underline{d}), \hat{v}(d))$ consumes a positive quantity below their satiation quantity d.*

Because of the positive affiliation, rationing on intensive margin is profitable to the seller because by having a consumer of type (v, d) with $v > \tau(d)$ consuming a quantity q below their satiation quantity, it improves the conditional distribution of per-unit value v at quantity q, allowing the seller to set a higher marginal price at q. Example 1 illustrates this effect with a simple example.

Example 1. *Let the type space be $\{(v_i, d_i)_i\} = \{(0.25, 0.5), (0.75, 1), (2, 1)\}$ where each type is drawn uniformly from the type space. Consumer type with higher per-unit value also has higher satiation quantity. There is one consumer type (type 1) that has a satiation quantity of* 0.5 *units and two consumer types (type 2 and 3) that have a satiation quantity of* 1 *unit. Now consider two menus for the seller. Menu 1, the seller sells a* 0.5*-unit bundle to type 1, and a* 1*-unit bundle to*

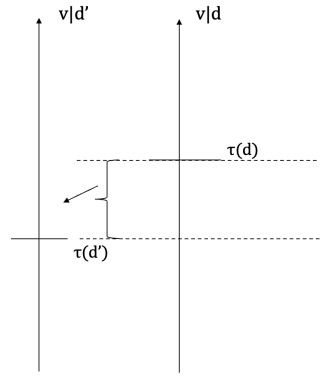


Figure 1: Illustration of the below satiation quantity consumption.

Notes: Consumers with higher satiation quantity will optimally consume a lower quantity. The two vertically lines represent the per-unit value v of the consumers of two different satiation quantity $d' < d$. Some consumers of values around the marginal price $\tau(d)$, shown as between the dashed lines, will consume a quantity below their satiation quantity d.

type 2 and 3 with profit

$$\frac{1}{3} \times \underbrace{(0.5 \times 0.25)}_{\substack{\textit{max. price} \\ \textit{for the 0.5-unit bundle}}} + \frac{2}{3} \times (0.5 \times 0.25 + 0.5 \times \underbrace{0.75}_{\substack{\textit{max. marginal price} \\ \textit{for the additional 0.5 units}}}) = 0.375.$$

Menu 2, the seller sells a 0.5*-unit bundle to type 2, and a* 1*-unit bundle to and 3 with profit*

$$\frac{1}{3} \times \underbrace{(0.5 \times 0.75)}_{\substack{\textit{max. price} \\ \textit{for the 0.5-unit bundle}}} + \frac{1}{3} \times (0.5 \times 0.75 + 0.5 \times \underbrace{2}_{\substack{\textit{max. marginal price} \\ \textit{for the additional 0.5 units}}}) \approx 0.583 > 0.375.$$

In menu 2, type 2 (with $v_2 = 0.75$ and $d_2 = 1$) chooses the 0.5-unit bundle, even though her satiation quantity is 1. In other words, she under-consumes relative to satiation. This downward pooling has two positive effects on seller profit. First, it improves the conditional distribution of per-unit values at quantity 0.5 -rather than only the low-value type with $v_1 = 0.25$, there are two types with values $v_1 = 0.25$ and $v_2 = 0.75$. The improved conditional distribution of per-unit values at quantity 0.5 allows the seller to set a higher price for the 0.5-unit bundle. Second, it also improves the conditional distribution of per-unit values at quantity 1 -rather than having two types with values $v_2 = 0.75$ and $v_3 = 2$, there is one type with value $v_3 = 2$. This allows the seller to set a higher marginal price for the additional 0.5 unit, that is, 2 rather than 0.75.

2.1.1 The case of a single-dimensional type

We now contrast our result with the canonical result of Maskin and Riley (1984), which shows that quantity discounts are almost everywhere optimal in the standard single-dimensional non-linear pricing problem. To that end, we let the consumer's satiation quantity d be a non-decreasing function of their valuation v such that v and d are perfectly positively correlated and the consumer has effectively a single-dimensional type.

The canonical result of Maskin and Riley (1984) shows that quantity discounts are almost everywhere optimal in the standard single-dimensional non-linear pricing problem. We now show that this is not true. To that end, we let the consumer's satiation quantity d to be a non-decreasing function of their valuation v such that v and d are perfectly positively correlated and the consumer has effectively a single type.

Quantity premium can be optimal in a simple single-dimensional type setting. The conditions about consumer preferences in Maskin and Riley (1984) rules out an important relationship between consumer's valuation and satiation.

And quantity premium is optimal in the setting when v and d are perfectly positively correlated such that the consumer has effectively a single-dimensional type. By doing so, we show that quantity premium is not a result of added dimensionality.

When $d(\cdot)$ is non-decreasing, consumers with a higher type—represented here by a higher per-unit value v—have a greater marginal utility across all quantities. Panel (a) of Figure 2 provides a visualization of an example of these consumer utilities. Importantly, the single-crossing preferences assumption remains satisfied. Consequently, the problem can be solved using the standard envelope theorem approach. We formally present the resulting optimal mechanism in Proposition 2.

Proposition 2. *When $d(\cdot)$ is non-decreasing, the optimal mechanism has the allocation rule that uses the same cut-off type in a standard optimal auction, $\hat{v}$. In the optimal mechanism, for all $v < \hat{v}$, $q(v) = 0$; for all $v \geq \hat{v}$, $q(v) = d(v)$ and $T(v) = vd(v) - \int_{\hat{v}}^{v} d(x)dx$.*

See Appendix A.4 for the proof.

The optimal mechanism features a cut-off type, $\hat{v}$. This cut-off is, perhaps surprisingly, identical to that found in a standard optimal auction when all consumers have the same satiation quantity. The reason for this is that the seller's optimization problem remains linear with respect to the consumer's allocation. Therefore, the core principle of mechanism design persists: the seller allocates a consumer their satiation quantity $d(v)$ if and only if that consumer's virtual value is non-negative, that is, whenever $v \geq \hat{v}$.

Given $v > \hat{v}$, one can calculate the per-unit price the type $(v, d(v))$ pays

$$t(v) = \frac{T(v)}{d(v)} = \underline{v} + \int_{\underline{v}}^{v} 1 - \frac{d(x)}{d(v)} dx$$

For $v' > v \geq \hat{v}$,

$$\begin{aligned} t(v') = \frac{T(v')}{d(v')} &= \underline{v} + \int_{\underline{v}}^{v} \underbrace{1 - \frac{d(x)}{d(v')}}_{>1-\frac{d(x)}{d(v)}} dx + \underbrace{\int_{v}^{\underline{v}} 1 - \frac{d(x)}{d(v')}}_{\geq 0} dx \\ &\geq t(v). \end{aligned}$$

Therefore, the optimal mechanism exhibits quantity premium.

Corollary 2. *If $d(\cdot)$ is non-decreasing, then the optimal mechanism exhibits quantity premium.*

A useful way to understand the intuition behind the quantity premium result is to consider how the *marginal* price at each unit separates consumers of different types.

Let $T(q)$ be any pricing schedule that maps quantities to aggregate prices. For a consumer that is already purchasing q units, it is optimal for them to purchase $q + \Delta$ units only if the marginal price at q, $T(q + \Delta) - T(q)$, is less than their marginal utility, in this case, v, at q. Thus, the marginal price at q essentially acts as the cut-off type that determines which consumers find it optimal to consume at least q units. The set of marginal utilities at each q is $\{v : d(v) \geq q\}$. Let F_q denote the distribution of the marginal utilities at q. Crucially, F_q monotone likelihood ratio dominates $F_{q'}$ for any $q > q'$. It then follows that the optimal cut-off type, that is the marginal price, for each quantity q is increasing, explaining the quantity premium. Figure 2 provides an illustration for this intuition. In addition, Example 2 provides a simple two-type example.

Example 2. *Let the type space be $\{(v_i, d_i)_i\} = \{(1, 0.5), (2, 1)\}$ where (v, d) is uniformly drawn from its type space. The consumer type with higher per-unit value, $v_2 = 2$, also has higher satiation quantity, $d_2 = 1$. Given that the consumer of type 1 and 2 has satiation quantities* 0.5 *units and* 1 *unit respectively, the ideal menu consists of a* 0.5*-unit bundle targeting the consumer of type 1 and a* 1*-unit bundle targeting the consumer of type 2. To achieve this, the marginal price from* 0 *units to* 0.5 *units cannot exceed the marginal utility $v_1 = 1$ of the type 1 consumer, and the marginal price from* 0.5 *units to* 1 *unit cannot exceed the marginal utility $v_2 = 2$ of the type 2 consumer. Thus, the highest price the seller can charge for the* 0.5*-unit bundle is $0.5 \times v_1 = 0.5$. The highest* marginal *price that the seller can charge for the* 1*-unit bundle is $0.5 \times v_2 = 1$. That*

makes the highest price that the seller can charge for the 1*-unit bundle*

$$0.5 \times \underbrace{1}_{\textit{marginal price for the first 0.5-unit}} + 0.5 \times \underbrace{2}_{\textit{marginal price for the additional 0.5-unit}} = 1.5.$$

The optimal menu is thus $\{(t, q)\} = \{(0.5, 0.5), (1.5, 1)\}$*, featuring a quantity premium as the* 1*-unit bundle has a higher per-unit price of* 1.5 *per unit than the* 0.5*-unit bundle which has a per-unit price of* 1*.*

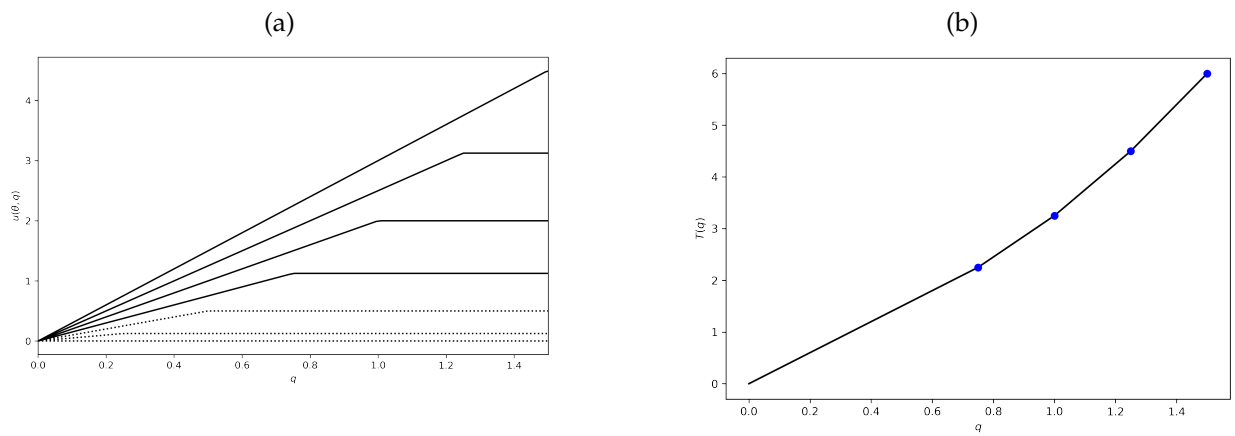


Figure 2: Panel (a): visualization of consumer utility curves $u(\theta, q)$ of different types θ as a function of quantity q when the maximum demand is increasing. Here $v \in \{0, 0.5, 1, 1.5, 2, 2.5, 3\}$ and $d(v) = 0.5v$. Consumers of higher type have higher marginal utilities for all quantities. Solid lines represent consumer types that are allocated with a positive quantity under the optimal pricing. Panel (b): the optimal price schedule for Panel (a). The cutoff type is $v = 1.5$. The gradient of the optimal price schedule at $q \geq 0.75$ corresponds to the marginal utility of the consumer whose maximum demand is q.

A central theme of this paper is that affiliation between per-unit value and satiation quantity enhances the effectiveness of quantity-based screening. Corollary 3 provides a direct manifestation of this principle: when satiation increases with valuation, the seller can set strictly higher per-unit prices than in the benchmark case where all consumers share the same satiation quantity.

Corollary 3. *If* $d(\cdot)$ *is increasing, the per-unit price in the optimal mechanism is strictly higher than* $\hat{v}$*, except for the marginal type* $\hat{v}$*.*

While our model allows for type-dependent satiation quantities, such variation need not stem from preferences alone. Type-dependent satiation quantities can arise from exogenous policy interventions. For example, governments sometimes use means-tested

programs that provide goods at no cost, or at substantially reduced prices, to lower-income consumers. Consider a government program that provides free or subsidized food, housing, or childcare to households below a certain income threshold. Once eligible consumers satisfy their basic needs through the public program, their satiation quantity for additional purchases in the private market lowers. Higher-income consumers, who are ineligible for such assistance, must rely entirely on private market purchases and thus have higher satiation quantities. This may lead satiation quantities in the private market to increase with consumer type (income). One interpretation of Corollary 3 is that consumers face higher per-unit prices in the private market following the introduction of means-tested provision.

2.2 Negative affiliation and quantity discount

When v and d are negatively affiliated, the conditional cut-off type $\hat{v}(q)$ is non-increasing in q. Unlike the positive affiliation case, the pricing schedule with marginal price at q equal to the conditional cut-off type $\hat{v}(q)$ no longer implements the cut-off types. This is because, given a non-increasing marginal price, a consumer of type $(v, d$ either consumes zero or d units. In particular, the consumer consumes d units as long as $\int_0^d \tau(q)dq \leq v \cdot d$. Thus, given a pricing schedule with $\tau(q) = \hat{v}(q)$, the cut-off type $\tilde{v}(q)$ for each quantity q satisfies that

$$\begin{aligned} q \cdot \tilde{v}(q) &= \int_0^q \hat{v}(x)dx \\ \tilde{v}(q) &= \frac{\int_0^q \hat{v}(x)dx}{q}. \end{aligned} \tag{1}$$

So $\tilde{v}(q) \neq \hat{v}(q)$ as long as $\hat{v}(q)$ is not a constant.[7]

The right-hand side of Equation (1) is the average price at q. This suggests that a pricing schedule that sets the average price at q equal to the conditional cut-off type $\hat{v}(q)$ could implement the cut-off type. However, a caveat arises: the implied total price $\hat{v}(q) \cdot q$ at each quantity q may be decreasing in q, which would violate incentive compatibility constraint. We therefore impose the following assumption to rule out this case.

Assumption 1. *$\hat{v}(d) \cdot d$ is non-decreasing in d.*

Under Assumption 1, the pricing schedule with average price at q equal to the conditional cut-off type $\hat{v}(q)$ implements the cut-off types and is optimal. Since $\hat{v}(q)$ is non-

[7] In fact, the implemented cut-off type $\tilde{v}(q) > \hat{v}(q)$ since $\hat{v}(\cdot)$ is non-increasing.

increasing in q, the optimal pricing exhibits quantity discounts. We summarize this result in Proposition 3.

Proposition 3. *If v and d are negatively affiliated and Assumption 1 holds, the optimal pricing exhibits quantity discount and the average price $t(q) = \hat{v}(q)$.*

Notably, unlike the positive affiliation case, rationing on the intensive margin is no longer profitable for the seller —there is insufficient mass of consumers with both high per-unit value and high satiation quantity to to generate the same improvement in the conditional distribution of per-unit values that the seller achieves with quantity premium under positive affiliation[8]. All participating consumers consume their satiation quantity. This is an immediate result from Proposition 3. We summarize it in Corollary 4

Corollary 4. *If v and d are negatively affiliated and Assumption 1 holds, the participating consumers are of types (v, d) that satisfies $v \geq \hat{v}(d)$ and all participating consumers consume their satiation quantity.*

Assumption 1 imposes further restrictions on the distribution of (v, d). We now focus on a special case of negative affiliation in which the satiation quantity $d(v)$ is non-increasing in per-unit value v. This allows us to examine the implications of relaxing Assumption 1 and fully characterize the optimal mechanism. Moreover, this single-dimensional case is of independent interest because consumer preferences violate the standard single-crossing condition.

2.2.1 The case of a single-dimensional type

When the satiation quantity $d(v)$ is decreasing in per-unit value v, the consumer's preference no longer satisfies the single-crossing condition. Figure 3 provides a visual comparison of the consumer utility curves that satisfy the single-crossing condition and the consumer utility curves corresponding to a decreasing satiation quantity.

The single-crossing condition ensures monotonicity in consumer types: the ranking of both utility and marginal utility among different types is preserved as quantity changes. This monotonicity property implies that, under any price schedule with non-negative marginal prices, higher-type consumers purchase weakly more than lower-type consumers. This sorting behavior explains why optimal price schedules feature strictly positive marginal prices. Suppose the optimal mechanism allocates q units to type θ and strictly more than q units to all types $\theta' > \theta$. Setting the marginal price at q just slightly higher than the

[8]See Example 1

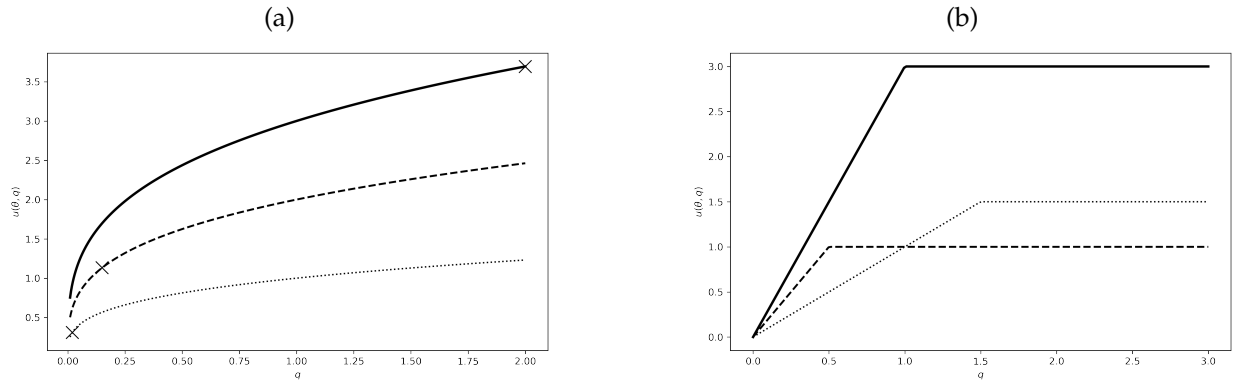


Figure 3: Panel (a): visualization of consumer utility curves $u(\theta, q)$ of different types θ as a function of quantity q that satisfies the single-crossing condition. Panel (b): visualization of consumer utility curves $u(\theta, q)$ of different types θ as a function of quantity q when the satiation quantity decreases in per-unit value.

Notes: In panel (a), consumers of higher type have higher utilities and marginal utilities for all quantities. In panel (b), the order of consumer utility is not the same across different quantities: the consumer with utility represented by dashed curve than the consumer with utility represented by the dotted curve for $q < 1$ but the order swaps once $q > 1$. The order of consumer marginal utility is not the same across different quantities: the solid curve consumer has higher marginal utility than the dotted curve consumer for $q < 1$ but the order swaps once $q > 1$.

marginal utility of type θ (which is strictly positive) does not screen out any intended purchasers: by the single-crossing condition, all types $\theta' > \theta$ have higher marginal utility at q than type θ, so they continue purchasing beyond q. Meanwhile, this positive marginal price extracts additional revenue from all types purchasing beyond q, relative to a zero marginal price.

When solving a standard problem where the consumer preference satisfies the single-crossing condition, incentive compatibility requires the price of the high quantity to be low enough such that a higher type gets weakly higher payoff choosing the high quantity than the low quantity. The single-crossing property further ensures that only the downward incentive compatibility constraints are binding. Hence, leaving sufficient surplus to higher types at higher quantities is sufficient for incentive constraints. One does not need to worry about the lower types deviating to a higher quantity bundle.

With satiation quantity decreasing in per-unit value, however, two important complications arise. First, a type with lower marginal utility at a lower quantity may have higher marginal utility at a higher quantity, so satisfying the downward incentive constraint alone is *no longer* sufficient; such a type may profitably deviate to a higher-quantity bundle. Example 3 provides a simple example to illustrate this. Second, the downward incentive constraint is relevant only if the quantity intended for the lower type (lower v)

is below the satiation quantity of the higher type (higher v). If instead the quantity intended for the lower type exceeds the satiation quantity of the higher type, then—given that higher quantities are associated with higher prices—the higher type would never choose that allocation, and the usual constraint of leaving sufficient surplus to higher types no longer applies.

Example 3. *Let the type space be* $\{(v_i, d_i)_i\} = \{(1.5, 1), (2, 0.5)\}$ *where* (v, d) *is uniformly drawn from the type space. Type 2 has a higher per-unit value,* $v_2 = 2$*, but a lower satiation quantity,* $d_2 = 0.5$.[9] *This preference profile violates the single crossing condition: the ranking of the marginal utility among the consumer types changes as quantity changes. In particular, type 2 has higher marginal utility for* $q < 0.5$*, while type 1 has higher marginal utility for* $q > 0.5$*. If one mechanically applies the standard envelope-theorem recipe—binding type 1's participation constraint at* $q = 0.5$ *and then binding type 2's incentive constraint—one obtains the menu* $\{(t, q)\} = \{(0.75, 0.5), (0.75, 1)\}$[10]*. However, this menu is* neither optimal nor incentive compatible*. Type 1 will optimally choose* $(0.75, 1)$ *since it provides higher utility.*

We characterize the optimal pricing in two steps. First, we approximate the satiation quantity function $d(\cdot)$ with n-step functions $d^n(\cdot)$. We show that the optimal pricing when the satiation quantity is the step function $d^n(\cdot)$ exhibits quantity discounts. this quantity-discount structure is preserved in the limit as the approximation step size approaches zero and $d^n(\cdot) \to d(\cdot)$. Second, using the established quantity-discount property, we derive the detailed form of the optimal allocation and the associated payment rule. We outline the key intuitions in the main text and provide the formal proof in the appendix.

For the intuition of the quantity discount, consider the following satiation quantity function that is a step function and non-increasing in the per-unit value $v \in [\underline{v}, \overline{v}]$

$$d(v) = \begin{cases} d_1, & v \in [\underline{v}, v^1) \\ d_2, & v \in [v^1, \overline{v}] \end{cases}$$

for some $0 < d_2 < d_1$. For cleaner presentation, let $U(v, T(v), q(v)$ denote the payoff of a consumer with per-unit value v given the allocation $q(v)$ and the payment $T(v)$. The incentive constraints of this problem can be decomposed into two parts:

[9] One can think of type 2 consumer as one that values Netflix at \$2 per hour but has only 30 minutes per day to watch and type 1 consumer as one that values Netflix at \$1.5 per hour but has 1 hour per day to watch.

[10] The highest feasible price is the utility that type v_1 obtains from consuming 0.5 units, namely $0.5 \times 1.5 = 0.75$. If the higher type, $v_2 = 2$, consumes this bundle $(0.5, 0.75)$, their payoff is 0.25. To ensure incentive compatibility, the price for the $q = 1$ bundle must therefore be at most $\min 1, 0.5 \times 2 - 0.25 = 0.75$. This yields the menu $(0.75, 0.5), (0.75, 1)$.

- (IC 1): for all $v, v' \in [\underline{v}, v_d)$ and for all $v, v' \in [v_d, \overline{v}]$,

$$U(v, T(v), q(v)) \geq U(v, T(v'), q(v'))$$

- (IC 2): for all $v \in [\underline{v}, v_d)$ and $v' \in [v_d, \overline{v}]$

$$U(v, T(v), q(v)) \geq U(v, T(v'), q(v'))$$

  and for all $v' \in [\underline{v}, v_d)$ and $v \in [v_d, \overline{v}]$

$$U(v, T(v), q(v)) \geq U(v, T(v'), q(v')).$$

The two sets of IC constraints can be further reduced. First, (IC 1) can be replaced by a local IC constraint by appealing to the envelope theorem. If (IC 1) is sufficient, then the optimal mechanism would have two cut-off types v_1 and v_2 such that

$$v_2 - \frac{1 - F(v_2)}{f(v_2)} = 0, \quad v_1 - \frac{1 - F(v^1)}{f(v_1)} = 0.$$

Second, (IC 2) can be replaced by a simple constraint: the aggregate payment $T(\cdot)$ is non-decreasing in the allocation $q(\cdot)$. The solution to the this problem has a simple form.

If $d_2 \cdot v_2 \leq d_1 \cdot v_1$, (IC 2) is not binding. The optimal mechanism is to offer d_1 units at a total payment of $d_1 v_1$ and d_2 units at a total payment $d_2 v_2$. The per-unit price for the larger bundle $d1$ is $v1$, which is lower than the per-unit price $v2$ for the smaller bundle $d2$ — i.e., the optimal menu exhibits a quantity discount. If $d_2 \cdot v_2 > d_1 \cdot v_1$, (IC 2) binds. It is no longer optimal to set a different price for d_1 units and d_2 units. The optimal solution is to set a fixed fee $T \in \arg\max_{T'} T' \cdot P(vd(v) \geq T')$, and consumers can consume their satiation quantity after paying the fixed fee T. This is effectively a quantity-discount mechanism because higher-demand types obtain more units for the same fee.

When we increase the number of steps in the satiation quantity function to n steps such that $d(v) = d_i$ for $v \in [v^{i-1}, v^i)$ for each $i \in \{1, \ldots, n-1\}$, the optimal mechanism has the same qualitative structure as in the two-step case. The cut-off types are now v_i's where for each i

$$v_i - \frac{F(v^i) - F(v_i)}{f(v_i)} = 0.$$

Note that $v_i < v_{i-1}$ for all i. For any $d_{i-1} > d_i$, there are two possibilities: either (IC 2) is

not binding, then the optimal per-uni prices for d_{i-1} and d_i are $v_i < v_{i-1}$; or (IC 2) binds, the optimal mechanism is a fixed fee for both $d_{i-1} > d_i$. In both scenarios, the optimal menu displays quantity discounts: per-unit prices do not increase with quantity.

As $n \to \infty$, $d^n(\cdot) \to d(\cdot)$ uniformly. The quantity discount feature remains in the solution. With linear utility and a non-increasing marginal price schedule, individual demand is bang-bang: a consumer with type v either purchases $d(v)$ or 0.

This bang-bang structure in allocation in turn pins down the shape of payments. Since the satiation quantity $d(v)$ is decreasing in v and the served types, consumers who receive a positive amount, receive their satiation quantity $d(v)$, the allocation $q(v)$ among all the served types is decreasing in v. Consequently, the payment function among served types, $T(v)$, is also decreasing in v.

Given this knowledge on the shape of the payment function, we consider an auxiliary problem: maximize expected revenue (i.e., captured consumer surplus) over all non-increasing payment functions, subject only to individual rationality. That is,

$$\overline{T}(v) \in \arg \max_{\hat{T}(\cdot) \text{ non-increasing}} \int_{\hat{T}(v) \leq v \cdot d(v)} \hat{T}(v) \, dF(v) \tag{2}$$

Given the solution $\overline{T}(v)$ to the auxiliary problem, the candidate payment function is

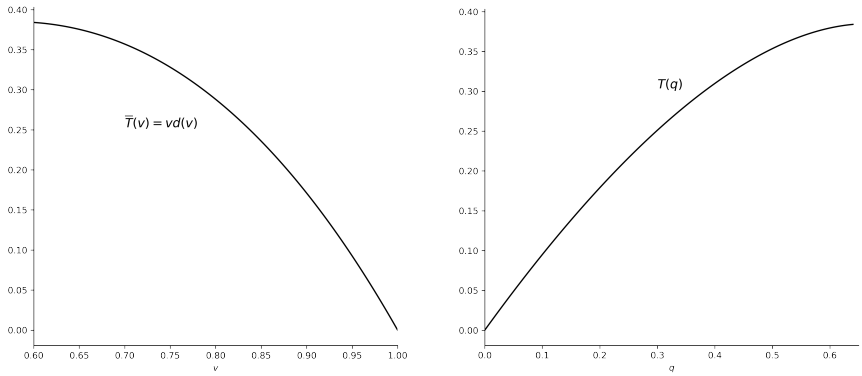
$$T(v) = \begin{cases} \overline{T}(v), & \overline{T}(v) \leq v \cdot d(v) \\ 0, & \overline{T}(v) > v \cdot d(v). \end{cases}$$

Equivalently, the candidate payment function is the largest (in the F-integral sense) function that lies below the envelope $v \cdot d(v)$ and is non-increasing for its non-zero segments.

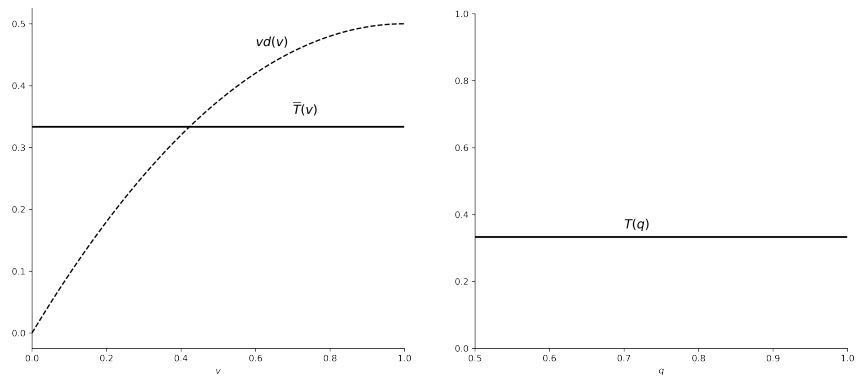
Two observations help characterize the solution to the auxiliary problem $\overline{T}(v)$ and the candidate payment function $T(v)$. First, if $v \cdot d(v)$ is non-increasing, then the $\overline{T}(v)$ equals $v \cdot d(v)$. Second, if $v \cdot d(v)$ is increasing, then the $\overline{T}(v)$ is a constant function. In fact, together with the segments where $T(v) = 0$, these are the only three kinds of segments of $T(v)$: 1) $T(v) = v \cdot d(v)$ for some non-increasing segments of $v \cdot d(v)$, 2) $T(v)$ is a constant for some increasing or non-monotone segments of $v \cdot d(v)$, and 3) $T(v) = 0$. To see why this is the case, consider any non-increasing function $h(v)$ that lies below $v \cdot d(v)$ for some interval $[v_1, v_2]$. Then one can construct another function $\overline{h}(v)$ that consists of only 1) and 2) and satisfies that $\overline{h}(v) \geq h(v)$ for all $v \in [v_1, v_2]$. Figure 4 provides visualizations of each of these different shapes of consumer's maximum surplus $vd(v)$, its associated candidate payment function $\overline{T}(v)$ and its optimal pricing schedule $T(q)$.

In contrast to the standard mechanism design approach—where we usually first find

(a) All-you-can-pay/Full surplus extraction



(b) All-you-can-eat/Zero marginal price



(c) Tiered pricing with an 'unlimited' option

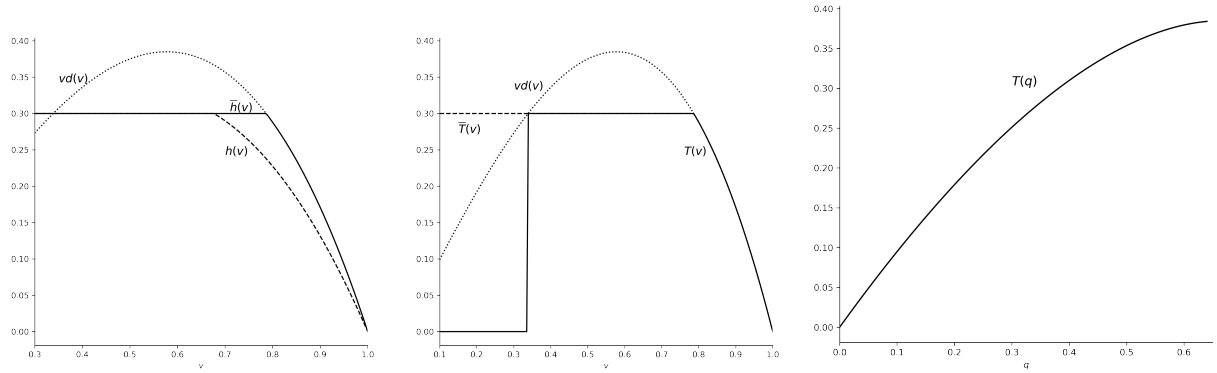


Figure 4: Examples of different shapes of consumer's maximum surplus and the associated optimal pricing schedule

Notes: Panel (a): consumer's maximum surplus $vd(v)$ is decreasing in v. The optimal $\overline{T}(v)$ coincides with $vd(v)$ (left). The optimal pricing schedule $T(q)$ is a smooth price schedule that varies continuous with quantity (right). Panel (b): consumer's maximum surplus $vd(v)$ is increasing in v. The optimal $\overline{T}(v)$ is a constant (left). The optimal pricing schedule $T(q)$ is also a constant (right), resembling the commonly observed all-you-can-eat, subscription pricing, and unlimited options. Panel (c): consumer's maximum surplus $vd(v)$ is non-monotone. On the right panel, given any non-increasing function, here $h(v)$ (dashed), another non-increasing function that consists of a constant and a segment equal to $vd(v)$, here $\hat{h}(v)$ (solid) improves the seller's revenue. On the middle panel, $\overline{T}(v)$ (dashed) is the solution to (2) and $T(v)$ (solid) is the optimal payment function. On the left panel, the optimal pricing schedule $T(q)$ consists of a segment of smooth pricing schedule that varies continuous with quantity, resembling the tired pricing structure seen in practice, and a constant segment, resembling an 'unlimited' option.

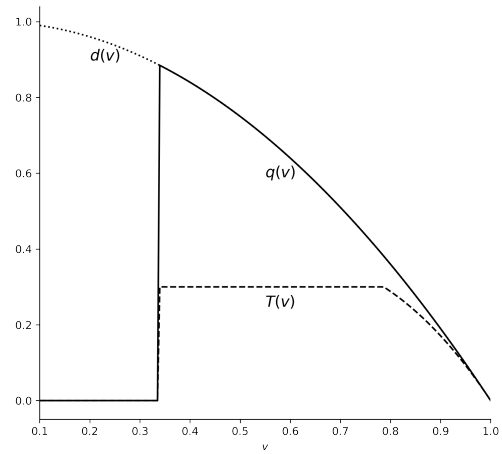


Figure 5: Allocation (solid line $q(v)$) determined by the candidate payment function (dashed line $T(v)$).

Notes: The dotted line $d(v)$ is the satiation quantity from the example in Figure 4 (d), and the dashed line $T(v)$ is the associated candidate payment function. The allocation coincides with the satiation quantity whenever $T(v) > 0$ and equals zero whenever $T(v) = 0$.

the optimal allocation and then use incentive compatibility to write down the payments—here we first have the candidate payment $T(v)$, then the candidate allocation function is determined by $T(v)$.

$$q(v) = \begin{cases} d(v), & T(v) > 0 \\ 0, & T(v) = 0. \end{cases}$$

Essentially, consumers with types v that is paying a positive amount, i.e., $T(v) > 0$, receive their satiation quantity $d(v)$ while others receive 0. Figure 5 provides a visual illustration on how allocation is determined by the payment function.

Now it remains to verify that the candidate allocation and payment functions satisfy global incentive compatibility. First note the price schedule $(q(v), T(v))_v$ features quantity discount: on the segments where $T(v)$ is a constant, the per-unit price $\frac{T(v)}{q(v)}$ is lower for higher $q(v)$; on the segments where $T(v) = vd(v)$, the per-unit price for a quantity $d(v)$ is v, which decreases in $d(v)$. Given the quantity discount, each consumer optimally purchase their satiation quantity conditional on that they can afford it: there is no incentive to purchase more since it implies higher payments and zero marginal utility and there is no incentive to purchase less since it implies higher per-unit price.

We state the optimal mechanism in Proposition 4. Figure 6 illustrates an example where the optimal mechanism has all of the three features.

Proposition 4. *When $d(\cdot)$ is decreasing, the optimal mechanism consists of a sequence of $(v_r)_{r\geq 0}$, with payment and allocation satisfies one of three conditions*

1. *(all-you-can-pay) $q(v) = d(v)$ and $T(v) = vd(v)$ for all $v \in [v_r, v_{r+1}]$. $vd(v)$ is non-increasing in $v \in [v_r, v_{r+1}]$.*

2. *(all-you-can-eat) $q(v) = d(v)$ and $T(v) = v_r d(v_r)$ for all $v \in [v_r, v_{r+1}]$. $vd(v)$ is increasing or non-monotonic in $v \in [v_r, v_{r+1}]$.*

3. *(non-serving) $q(v) = 0$ and $T(v) = 0$ for $v \in [v_r, v_{r+1}]$.*

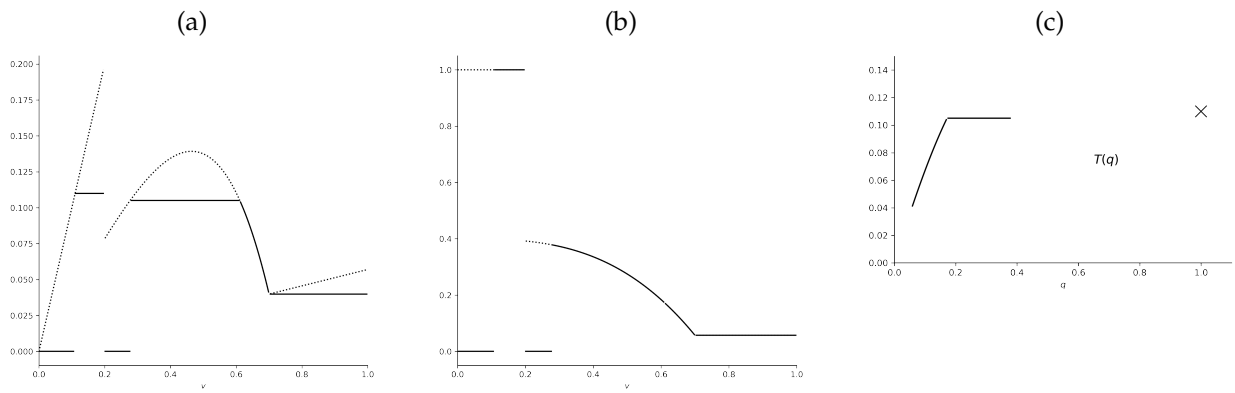


Figure 6: An example where the optimal mechanism has all of the three features

Notes: The optimal payment and allocation for $v \sim U[0,1]$ and $d(v) = 1$ for $v < 0.2$, $d(v) = 0.4 - v^3$ for $v \in [0.2, 0.7]$ and $d(v) = 0.057$ for $v > 0.7$. Panel (a): the optimal payment $T(v)$ (solid line) for each type and the maximum surplus $vd(v)$ (dotted line). Panel (b): the optimal allocation (solid line) and the satiation quantity $d(v)$ (dotted line).Panel (c): the optimal tariff $T(q)$ consists of first a smoothly varying price schedule, a flat fee of 0.105 for any quantity in $[0.172, 0.379]$, and an option to buy 1 unit at 0.11.

We now illustrate the economic intuitions and implications behind these three regions. In Section 3, we discuss when these results do and do not generalize generalize beyond the satiation model.

All-you-can-pay/full surplus extraction In the all-you-can-pay regions, $vd(v)$ is non-increasing. It is feasible and optimal to perfectly separate all the different types. Consumers consume their satiation quantity $d(v)$ and pay their *full surplus* $vd(v)$. Here, not only the order of the marginal utilities are not preserved across quantities, the order of the *utilities* are also not preserved across quantities. The change in the order of the utilities makes the screening costless and full surplus extraction possible. Within this region, for each quantity q, there exists a unique type that attains the highest utility for q. By pricing that quantity q exactly at this maximum value, the seller ensures that only this

specific type can afford the bundle. Applying this principle across all quantities in the region leaves each consumer type with exactly one affordable option, perfectly screening them and preventing any profitable deviation. Consequently, the individual rationality constraint binds for every type, leaving them with zero net utility. Example 4 provides a simple example that illustrates the mechanism at work.

Example 4. *Consider the type space* $\{(v_i, d_i)_i\} = \{(1.5, 1), (2, 0.5)\}$ *where each type* (v, d) *is uniformly drawn from the type space. The consumer types satisfies that type 2 consumer has a higher per-unit value, but a lower satiation quantity* and *a lower maximum surplus,* $v_2 \times d_2 < v_1 \times d_1$*. At* $q = 0.5$*, consumer of type 2 has the higher willingness to pay (utility of* 1*, versus* 0.75 *for consumer of type 1). At* $q = 1$*, the roles reverse: consumer of type 1 now has the higher willingness to pay (utility of* 1.5*, versus* 1 *for consumer of type 2). The optimal menu makes use of both the change in rank in utility and marginal utility of the two types. Consider* $\{(t, q)\} = \{(1, 0.5), (1.5, 1)\}$*. Type 1 consumer will choose the bundle* $(1.5, 1)$ *because the per-unit price of the other bundle* $(1, 0.5)$ *is* 2 *exceeding her per-unit value of* $v_1 = 1.5$*. Type 2 consumer will choose the bundle* $(1, 0.5)$ *because the price of the other bundle* $(1.5, 1)$ *is* 1.5 *exceeding her maximum surplus for a 1-unit bundle which is* 1*. Each consumer pay their maximum surplus and has their surplus fully extracted in the optimal menu.*

Within our model, the existence of a unique type that attains the highest utility for each q is sufficient for all-you-can-pay mechanism to be optimal. This can be generalized to other consumer preferences but with additional conditions on how 'separable' the types are. Section 3 provides examples for when all-you-can-pay mechanism is and is not optimal for more general consumer preferences.

Standard economic intuition holds that monopoly inefficiency stems from under-provision: a monopolist restricts quantity to screen out lower-type consumers and charge higher prices to remaining consumers. However, this intuition relies critically on preferences satisfying the single-crossing condition. It is useful to analyze the nature of monopoly inefficiency when the single-crossing condition no longer holds. Throughout most of this paper, we assume zero production costs, under which the first-best allocation provides every consumer with their satiation quantity—implying that a monopolist necessarily under-provides (weakly). That comparison is trivial. To generate more meaningful insights, we now introduce positive production costs.

The efficient monopolist The optimality of the all-you-can-pay mechanism illustrates an important insight about consumer preferences that violate the single-crossing condition. When the ranking of consumers' marginal utilities (and utilities) is not preserved

across quantities, quantity becomes an even more powerful screening device. Specifically, when the maximum surplus function $vd(v)$ is decreasing in v, quantity screening is so effective that the monopolist can extract full consumer surplus under zero production costs. In this case, the monopolist achieves first-best efficiency.

Remarkably, efficiency can persist even with positive production costs. To formalize this result, we introduce the aggregate allocation function:

$$Q(v) := \int_v^{\overline{v}} d(x)dF(x),$$

which represents the total satiation quantity across all consumer types from v to $\overline{v}$. Given a quantity constraint $\hat{Q}$, the first-best allocation prioritizes consumers by their per-unit valuation (from highest to lowest) and serves each consumer their satiation quantity until the constraint binds. Specifically, there exists a marginal consumer type $\hat{v}$ such that all consumers of type $v > \hat{v}$ receive their satiation quantity, the marginal consumer of type $\hat{v}$ receives the residual amount, and consumers of type $v < \hat{v}$ receive nothing. The marginal type $\hat{v}$ satisfies

$$\hat{Q} \in [\lim_{v \to \hat{v}+} Q(v), \lim_{v \to \hat{v}-} Q(v)].$$

If $Q(v)$ is continuous in v, for any quantity constraint $\hat{Q}$, the marginal type $\hat{v}$ is simply $Q^{-1}(\hat{Q})$ and the marginal type also receives their satiation quantity. Given any quantity constraint $\hat{Q}$, the monopolist's optimal revenue is exactly the consumer surplus under the first best allocation. So the monopolist's payoff perfectly aligns with the first best. The monopolist is efficient. We state the result in Proposition 5.

Proposition 5. *If the consumer's maximum surplus $vd(v)$ decreases in v and $Q(v)$ is continuous in v, the monopoly provision is always equal to the first best quantity.*

Monopoly over-provision When $Q(v)$ is discontinuous in v, monopoly revenue need not align with first-best consumer surplus. To illustrate, consider a quantity constraint $\hat{Q}$ at a discontinuity point of $Q(v)$, so that $Q(v) \neq \hat{Q}$ for all v. The first-best allocation then assigns the marginal type $\hat{v}$ a quantity q strictly below their satiation quantity $d(\hat{v})$. The monopolist cannot implement this first-best allocation. Two cases demonstrate why: Case 1: If $q \leq d(v)$ for some $v > \hat{v}$, the monopolist cannot both offer q units at a per-unit price $\hat{v}$ and $d(v) = q$ units at a per-unit price $v > \hat{v}$. Case 2: If $q > d(v)$ for all $v > \hat{v}$ but $q\hat{v} < \lim_{v \to \hat{v}^+} vd(v)$, then if the monopolist offer q units at a per-unit price $\hat{v}$, some consumers with type $v > \hat{v}$ prefer to purchase $q > d(v)$ units at a total price $q\hat{v}$ than purchasing $d(v)$

units at a total price $d(v)v > q\hat{v}$, again violating incentive compatibility. Consequently, monopoly revenue at $\hat{Q}$ falls strictly below first-best consumer surplus. However, at some larger quantity $\hat{Q}' > \hat{Q}$ where $Q(v) = \hat{Q}'$ for some v (i.e., at a continuity point), monopoly revenue again equals first-best surplus. Importantly, the increase in monopoly revenue from $\hat{Q}$ to $\hat{Q}'$ exceeds the corresponding increase in first-best surplus. This wedge creates the possibility of over-provision: if the marginal cost of expanding production from $\hat{Q}$ to $\hat{Q}'$ lies between the first-best surplus gain and the monopoly revenue gain, the monopolist will choose $\hat{Q}'$, exceeding the first-best quantity $\hat{Q}$. Proposition 6 formalizes this result. Example 5 offers a simple example when the monopolist overprovides.

Proposition 6. *If the consumer's maximum surplus $vd(v)$ decreases in v and $Q(v)$ is not continuous in v, there exists a marginal cost function such that the monopoly provision is always weakly higher than the first best.*

Example 5. *Consider the type space $\{(v_i, d_i)_i\} = \{(1.5, 1), (2, 0.5)\}$ where each (v, d) is uniformly drawn from the type space. Let the incremental cost of increasing production from 0.5 unit to 1 units be $c > 0$. The maximum social welfare that can be obtained from 1 units of the good is $\frac{1}{2} \times 0.5 \times 2 + \frac{1}{2} \times 1 \times 1.5 = 1.25$; while the maximum social welfare from 0.5 units is $\frac{1}{2} \times 0.5 \times 2 + \frac{1}{2} \times 0.5 \times 1.5 = 0.875$. Thus supplying an extra 0.5 units raises social welfare by $1.25 - 0.875 = 0.375$. By contrast, the monopoly revenue rises by more: the monopoly revenue at 1 units is $\frac{1}{2} \times 0.5 \times 2 + \frac{1}{2} \times 1 \times 1.5 = 1.25$, whereas the monopoly revenue at 0.5 units is 0.75, so the monopolist's revenue gain from increasing output is 0.5! For $0.375 < c < 0.5$, the socially efficient provision is 0.5 units but a profit-maximizing monopoly will supply 1 unit.*

All-you-can-eat/Zero marginal price In the all-you-can-eat regions, $vd(v)$ is not monotonically decreasing. In sharp contrast to the all-you-can-pay region, it is no longer optimal to separate types. Instead, the optimal mechanism pools all types by offering a single bundle. This is driven by a fundamental conflict: the types with the highest total utility are now the same types with the lowest marginal utility. If there is a price differential for a larger quantity, consumers with low marginal utility are less willing to pay for additional quantity, so they will naturally choose the smaller bundle; however, in this region, these are precisely the consumers with the highest overall willingness to pay. Any price differential thus creates a perverse incentive, causing the seller's most valuable customers to purchase the cheaper option. This mechanism works against revenue generation. The only way to resolve this is to eliminate the incentive to self-select, which means setting the price differential to zero. Consequently, offering a single all-you-can-eat bundle, or equivalently setting a zero marginal price, is optimal. Example 6 provides a simple example to illustrate the mechanism.

Example 6. *Let the type space be* $\{(v_i, d_i)_i\} = \{(0.8, 1), (2, 0.5)\}$ *where each type* (v, d) *is drawn uniformly from the type space. The consumer types satisfies that higher per-unit value consumer* $(v_2 = 2)$ *has a lower satiation quantity* $(d_2 = 0.5)$ *but a higher maximum surplus* $v_2 \times d_2 > v_1 \times d_1$*. Ideally, the seller sells a 1-unit bundle to type 1 consumer whose satiation quantity is 1 and a 0.5-unit bundle to type 2 consumer whose satiation quantity is 0.5. Since type 2 consumer has a higher maximum surplus, the seller would want to make the 0.5-unit bundle for type 2 consumer as expensive as possible. However, there is no incentive compatible way to make the 0.5-unit bundle more expensive than the 1-unit bundle – otherwise type 2 consumer can purchase the 1-unit bundle. The highest price that seller can charge the 0.5-unit bundle is the price of the 1-unit bundle. This is the key intuition for zero marginal price. In this example, setting a fixed fee of* 0.8 *and allowing the consumer to consume any quantity is optimal.*

If $vd(v)$ is nowhere decreasing, then by Proposition 4, the optimal mechanism has only the all-you-can-eat and non-serving region. Corollary 5 states this result. Section 3.1.1 extends the optimality of zero marginal price to setting beyond the satiation framework.

Corollary 5. *If* $d(\cdot)$ *is decreasing and* $vd(v)$ *is nowhere decreasing, the optimal mechanism is all-you-can-eat for a fixed price* $\tilde{p}$ *where* $\tilde{p} \in \arg\max_p p \cdot \mathbb{P}(vd(v) \geq p)$.

Consumer welfare under zero marginal price When the marginal price for a good is zero, intuition suggests that high-quantity consumers benefit most: they obtain more units for free and therefore earn greater surplus. However, this logic considers marginal pricing in isolation while neglecting the consumer preference structure that induces a profit-maximizing seller to set a zero marginal price. In our model, when zero marginal pricing is optimal, consumers' per-unit valuations v are strongly negatively correlated with their satiation quantities d. Under the optimal mechanism, participating consumers consume their satiation quantities, and consumers with high satiation quantities have lower maximum surplus because of their lower per-unit valuations. As a result of this lower maximum surplus, under the optimal mechanism—which charges a fixed fee—these consumers obtain lower surplus net the fixed fee.Proposition 7 formally states the result.

Proposition 7. *If* $d(v)$ *is decreasing and* $vd(v)$ *is increasing, under the optimal mechanism, consumers who purchase higher quantities have lower surplus.*

The optimality of zero marginal pricing extends beyond the satiation model. Section 3.1.1 shows that when the single-crossing condition is reversed—so that consumers with higher utilities have lower marginal utilities—zero marginal pricing remains optimal. This connects the canonical single-crossing assumption to the absence of optimal

zero marginal pricing in canonical models. Zero marginal price is prevalent in practice: buffet restaurants, cruises, seasonal ski passes, AI chatbot subscriptions, streaming services, and fitness gyms all employ variants of this model. Corollary 5 and Section 3.1.1 provide a rationale for the optimality and prevalence of such pricing: when consumers who value the product most also experience sharply diminishing returns to quantity, pooling all types with a single menu dominates separating menus.

Non-monotonic allocation When satiation quantity decreases in v, the relationship between consumer type and maximum surplus can be non-monotonic: consumers with intermediate per-unit valuations may have the lowest maximum surplus $vd(v)$, making them the true "low types" who are excluded under the optimal mechanism. Consequently, the optimal allocation is not monotonic in v—both low-valuation and high-valuation consumers may be served while middle-valuation consumers are excluded. This non-monotonicity offers a theoretical foundation for "missing middle" phenomena, where markets fail to serve intermediate consumers despite serving those at both extremes.

Example 7 illustrates this.

Example 7. *Consider $v \sim U[0,1]$, $d(v) = 1$ for all $v < 0.2$ and $d(v) = 0.3$ for all $v \geq 0.2$. The maximum surplus $vd(v)$ (dotted line in Figure 7) is high for $v \in [0.2, 0.4]$ and for $v > 0.5$ and low for $v < 0.2$ and $v \in [0.4, 0.5]$. The optimal mechanism is to charge 0.2 for 1 unit and 0.15 for 0.3 units. Consumers with $v \in [0.4, 0.5]$ do not consume.*

3 Extensions

Our model makes a simplifying assumption that consumers have a satiation quantity beyond which marginal utility becomes zero and below which utility is linear. In this section, we discuss the extent to which our results generalize beyond this utility specification. For completeness, we derive the optimal mechanism when the two dimensions, v and d, are independent.

3.1 General utility functions

To extend beyond our satiation model, we consider a consumer with single-dimensional type θ and utility function $u(\theta, q)$ from consuming quantity q. We let $\bar{q}$ denote an exogenous upper bound on quantity. We examine whether two key results extend to this

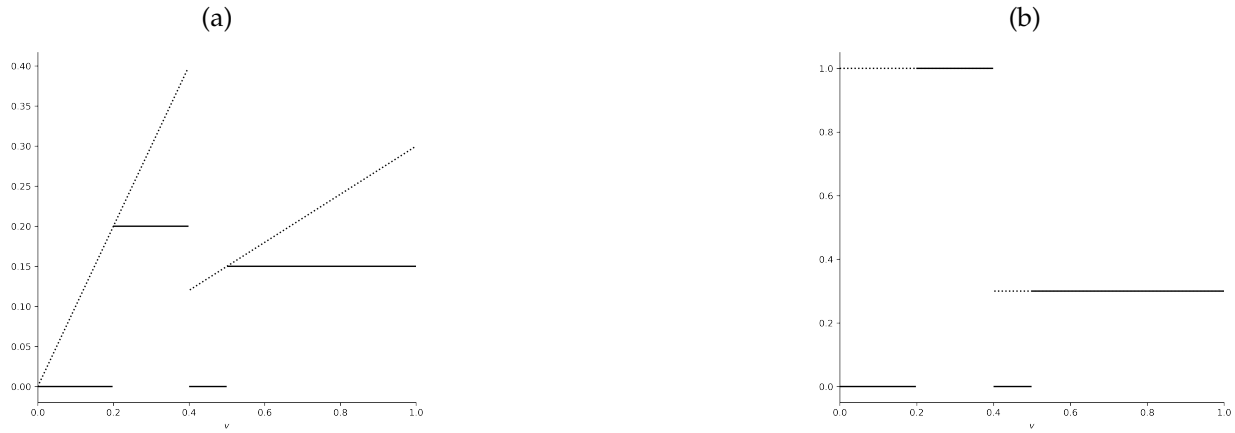


Figure 7

Notes: The optimal payment and allocation for Example 7. Panel (a): the optimal payment (solid line) and the maximum surplus $vd(v)$ (dotted line). Panel (b): the optimal allocation (solid line) and the satiation quantity $d(v)$ (dotted line). The optimal allocation first increases at $v = 0.2$, then decreases to 0 at $v = 0.4$, and then increases again at $v = 0.5$.

general setting: the optimality of zero marginal pricing and the optimality of full surplus extraction.

3.1.1 Optimality of zero marginal price and non-single crossing

In the main body of the paper, we discussed how the single-crossing condition leads to strictly positive marginal pricing: a positive marginal price increases revenue without unnecessarily screening off consumers when preferences satisfy single-crossing, since higher utility consumers also have higher marginal utility. In Section 2.2, we showed that zero marginal price is optimal when higher utility consumers (in the satiation model context, those with higher maximum surplus) have *lower* marginal utility (in the satiation model context, lower per-unit value). We now demonstrate that this result extends beyond the satiation model and provide a sufficient condition on the utility function $u(\theta, q)$ for the optimality of zero marginal pricing. The intuition is straightforward. When higher types (θ) have higher utility ($u_\theta(\theta, q) \geq 0$), they also have higher willingness to pay. Consequently, the seller would ideally want higher types to purchase larger quantities. However, when single-crossing is violated such that higher types have lower marginal utility ($u_{q\theta}(\theta, q) \leq 0$), any quantity-based screening requires setting a strictly positive marginal price. Given any positive marginal price, lower-type consumers will self-select into larger quantities because they have higher marginal utility. This conflicts with the seller's objective of inducing higher types to consume more. Hence, offering a single-item menu

at the quantity upper bound $\overline{q}$ is optimal. One implementation of this mechanism is to set an entry fee $\tilde{p}$ as in Proposition 8 and allow consumers to consume all they want. Proposition 8 formalizes this result.

Proposition 8. *If $u_\theta(\theta, q) \geq 0$ and $u_{q\theta}(\theta, q) \leq 0$ for all q and θ, then all-you-can-eat for a fixed price $\tilde{p}$ where $\tilde{p} \in \arg\max p\mathbb{P}(u_\theta(\theta, \overline{q}) \geq p)$ is optimal.*

Figure 8 illustrates this with an example when consumer preferences satisfy the conditions in Proposition 8.

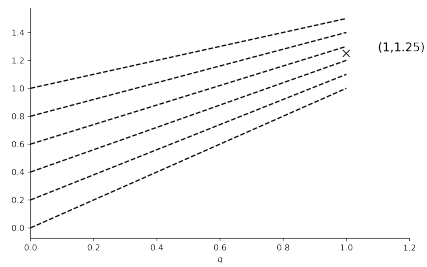


Figure 8: An example for Proposition 8

Notes: In this example, $\theta \sim U[0,1]$, $\overline{q} = 1$ and $u(\theta, q) = \theta + (1 - 1/2\theta)q$. Consumers with higher types θ have higher utilities for each q but lower marginal utility for each q. The optimal mechanism here is a fixed fee of 1.25 that allows each consumer to consume up to 1 unit.

3.1.2 When full consumer surplus extraction is feasible but not optimal

In the satiation model, we show that when consumers with higher per-unit value have lower maximum surplus (due to very low satiation quantities), the seller can achieve full consumer surplus extraction. The key insight is that for each quantity q, there exists a unique type that attains the highest utility at that quantity. The seller can set the price for the q-unit bundle equal to that type's utility, leaving each type with exactly one affordable option. However, perhaps counterintuitively, full consumer surplus extraction is not always optimal. Consider the utility function $u(\theta, q) = \theta q^\theta$, where type θ represents the consumer's elasticity. Case 1 (Figure 9, Panel (a)): Let there be two types $\theta \in \{0.5, 0.6\}$. The all-you-can-pay menu that perfectly separates the two types is $\{(T, q)\} = \{(0.162, 0.2), (1, 0.6)\}$, yielding revenue of 0.8. However, the optimal menu pools both types: a single bundle $(T, q) = (1, 0.5)$ yields revenue of 1.

Case 2 (Figure 9, Panel (b)): Let the two types be $\theta \in \{0.5, 1\}$. The all-you-can-pay menu is $\{(T, q)\} = \{(0.25, 0.25), (1, 1)\}$, yielding revenue of 1.25—the maximum achievable revenue. The alternative pooling menu $\{(T, q)\} = \{(1, 0.5)\}$ yields only 1.

The key difference between the two cases is the degree of type separability. In Case 2, the types are sufficiently separable: the utility curve for $\theta = 0.5$ (which is higher at lower quantities) flattens out quickly at higher quantities relative to $\theta = 1$ (which is higher at higher quantities). The seller gains sufficiently from quantity-based screening to justify separation. In Case 1, the types are not sufficiently separable: the utility curve for $\theta = 0.5$ does not flatten out quickly relative to $\theta = 0.6$. The revenue loss from selling less to type $\theta = 0.5$ outweighs the gain from perfect price discrimination, making pooling optimal.

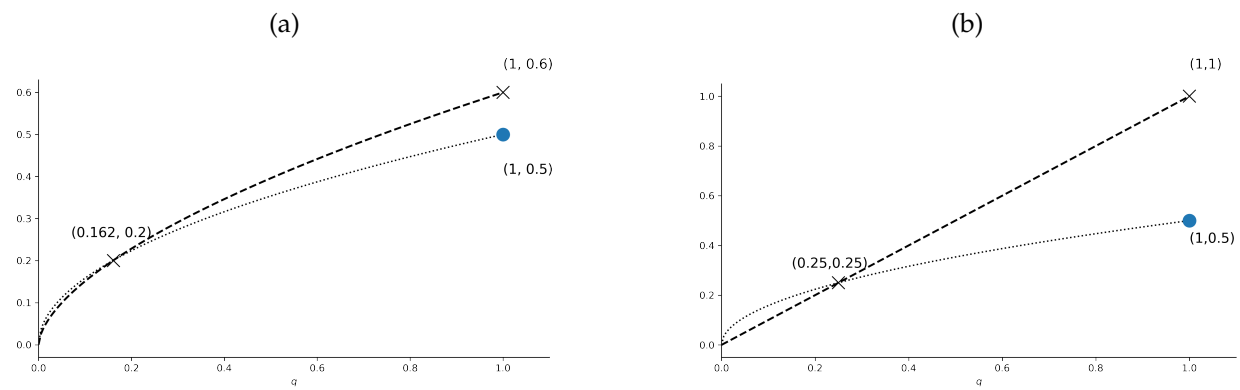


Figure 9: When all-you-can-pay mechanism is (b) and is not (a) optimal.

3.2 Independent v and d

When d and v are independently distributed, the conditional distribution of v given any satiation quantity d remains unchanged. With slight abuse of notation, let $F(\cdot)$ and $f(\cdot)$ denote the cumulative distribution function and probability density function of v, respectively, and let $\hat{v}$ denote the optimal cutoff type defined by

$$\hat{v} - \frac{1 - F(\hat{v})}{f(\hat{v})} = 0.$$

Under independence of v and d, the seller gains no additional information about the consumer's per-unit valuation v from observing the quantity consumed. Consequently, the optimal mechanism reduces to charging a uniform unit price $\hat{v}$.

Proposition 9. *When d and v are independent, the optimal pricing scheme is linear, with average price $t(q) = \hat{v}q$ for all q.*

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A Proofs

A.1 Proof of Lemma 2

We need to show that $\tau'(q) \geq 0$ implies that $t'(q) \geq 0$ and $\tau'(q) \leq 0$ implies that $t'(q) \leq 0$. For all q,

$$t(q)q = \int_0^q \tau(x)dx.$$

If $\tau'(q) \geq 0$, then $t(q)q = \int_0^q \tau(x)dx \leq \tau(q)q$. So $t(q) \leq \tau(q)$ which implies that $t'(q) \geq 0$. On the other hand, if $\tau'(q) \leq 0$, then $t(q)q = \int_0^q \tau(x)dx \geq \tau(q)q$. So $t(q) \geq \tau(q)$ which implies that $t'(q) \leq 0$.

A.2 Proof of Lemma 3

Given any pricing schedule $t(\cdot)$, a consumer with type $\theta = (v, d)$ solves the following problem

$$\max_q \quad v \cdot \min(q, d) - t(q)q.$$

Let $h(\theta, q) = v \cdot \min(q, d) - t(q)q$. The consumer's optimal choice is thus $q(\theta) \in \arg\max_q h(\theta, q)$. Let the partial order $>$ on Θ be such that $(v, d) > (v', d') := \theta'$ iff $v > v'$ and $d > d'$. Let $q' < q$. Assume that $h(\theta', q) \geq h(\theta', q')$, then

$$\begin{aligned}
h(\theta, q) - h(\theta, q') &= v \cdot \min(q, d) - t(q)q - \Big(v \cdot \min(q', d) - t(q')q'\Big) \\
&\geq v \cdot \min(q, d') - t(q)q - \Big(v \cdot \min(q', d') - t(q')q'\Big) \\
&> v' \cdot \min(q, d') - t(q)q - \Big(v' \cdot \min(q', d') - t(q')q'\Big) \\
&= h(\theta', q) - h(\theta', q') \geq 0.
\end{aligned}$$

So $h(\theta, q)$ has strict single-crossing property in (θ, q). By Theorem 4 in Milgrom and Shannon (1994), $q(\theta)$ is non-decreasing in θ.

A.3 Proof of Proposition 1

First, by Lemma 1, $\hat{v}(q)$ and hence $\tau(q)$ is non-decreasing in q. If $\tau(\cdot)$ is optimal, then the optimal pricing exhibits quantity premium. We show that $\tau(\cdot)$ is optimal.

Given $\tau(\cdot)$, a consumer with type (v, d) solves the following problem

$$\max_{q} \quad v \cdot \min\{d, q\} - \int_0^q \tau(x)dx.$$

Since the marginal price is non-negative, it is never optimal for a consumer to purchase $q > d$. So the consumer's problem is equivalent to

$$\max_{q:q \leq d} \quad vq - \int_0^q \tau(x)dx.$$

Let $\lambda > 0$ be the Lagrange multiplier on the constraint that $q \leq d$. Then the following are the necessary conditions on the solution

$$v = \tau(q) + \lambda,$$
$$0 = \lambda(q - d).$$

The solution $q = \min\{d, \tau^{-1}(v)\}$ satisfies the necessary condition and the second order condition since $\tau(\cdot)$ is non-decreasing. Notably, $q = \tau^{-1}(\hat{v}(q))$. Since $\tau(\cdot)$ is non-decreasing, all consumers of type (v, d) with $v \geq \hat{v}(q)$ and $d \geq q$ purchases at least q units. The marginal price $\tau(q) = \hat{v}(q)$ for all q implements the marginal cut-off types $\hat{v}(q)$.

A.4 Proof of Proposition 2

In any optimal mechanism, $q(v) \leq d(v)$ for all v because setting any $q(v) > d(v)$ worsens IC constraints for all $v' \neq v$ and does not improve revenue from the type v. Given $q(v) \leq d(v)$ for all v, the seller's problem is the same as the optimal auction problem in Myerson (1981) with the additional constraint that $q(v) \leq d(v)$ for all v. So the seller's problem can be re-written as

$$\max_{q} \int q(v)\phi(v)dF(v)$$

subject to $q(v)$ is non-decreasing in v and $q(v) \leq d(v)$ for all v. The mechanism in Proposition 2 yields the maximum revenue without the monotonicity constraint. Since $d(v)$ is

non-decreasing in v and $q(v) = d(v)$ satisfies the monotonicity constraint, the mechanism in Proposition 2 is also feasible to the original problem and hence optimal.

A.5 Proof of Proposition 3

Since $t(\cdot)$ is non-increasing and the aggregate price $t(q)q$ at q is non-decreasing in q under Assumption 1, a consumer of type (v, d) purchases d units if $t(d) \cdot d \leq v \cdot d$, and 0 units otherwise. So all consumers of type (v, d) with $v \geq t(d) = \hat{v}(d)$ purchases d units and $v < t(d) = \hat{v}(d)$ purchases 0 units. Hence the pricing schedule with average price $t(q) = \hat{v}(q)$ for all q implements the marginal cut-off types $\hat{v}(q)$ and is optimal. The optimal pricing exhibits quantity discount.

A.6 Proof of Proposition 4

We begin the proof by showing that the optimal mechanism exhibits price discount. Towards that end, we first construct a sequence of step functions $d^n(\cdot)$ that converges to $d(\cdot)$. Second let $\mathcal{C}$ be the correspondence that maps $d^n(\cdot)$ to the set of IR and IC mechanisms given the satiation quantity is $d^n(\cdot)$. We show that $\mathcal{C}$ is continuous. Then by Berge's maximum theorem, the optimal solution to the problem with satiation quantity $d^n(\cdot)$ converges to the optimal solution to the problem with satiation quantity $d(\cdot)$ and thus the optimal solution to the problem with satiation quantity $d(\cdot)$ has price discount.

For every $n \geq 1$, we divide the per-unit value space into 2^n equal intervals: $[v_0, v_1]$, $(v_1, v_2], \ldots, (v_{2^n-1}, v_{2^n}]$, where $v_0 = \underline{v}$, $v_{2^n} = \overline{v}$, $v_k = \underline{v} + \frac{k}{n}(\overline{v} - \underline{v})$. For each $v \in (v_{k-1}, v_k]$, let $d^n(v) = d(v_k)$. So $d^n \to d$ uniformly.

To show that $\mathcal{C}$ is upper hemicontinuous, let $(q^n, T^n)_n$ be a convergent sequence and let $(\tilde{q}, \tilde{T}) = \lim_{n\to\infty}(q^n, T^n)$, we show that $\in \mathcal{C}(d)$. Assume not. Then there exists types $v, v' \in [\underline{v}, \overline{v}]$ such that

$$v \min\{d(v), q(v)\} - T(v) < v \min\{d(v), q(v')\} - T(v').$$

Let N be such that $d^N(v) = d(v)$. Since $(\tilde{q}, \tilde{T}) = \lim_{n\to\infty}(q^n, T^n)$, there exists $N' > N$ large enough such that $(q^{N'}(v), T^{N'}(v))$ and $(q^{N'}(v'), T^{N'}(v'))$ is close enough to $(q(v), T(v))$ and $(q(v'), T(v'))$. Then the mechanism $q^{N'}, T^{N'}$ is also not IC. A constradiction.

To show that $\mathcal{C}$ is lower hemicontinuous, take any $(q, T) \in \mathcal{C}(d)$, we construct a convergent sequence of (q^n, T^n) and show that $(q^n, T^n) \in \mathcal{C}(d^n)$. Given (q, T), for any n and $k \leq 2^n$, let $M = \{(q(v_k), T(v_k))\}$ be the set of menus that consist only bundles offers to $(v_k)_{k\leq 2^n}$. For each $v \in [\underline{v}, \overline{v}]$, let $(q^n(v), T^n(v)) \in \arg\max_{(q,T)\in M} v \min(d^n(v), q) - T$.

Clearly, $(q^n, T^n) \to (q, T)$ as $n \to \infty$ and satisfies IR and IC.

Given the price discounts, consumers of per-unit value v either purchase their satiation quantity $d(v)$ or purchase nothing. So we search among the price schedules that result in allocations $q(v) \in \{0, d(v)\}$ for all $v \in [\underline{v}, \overline{v}]$. Thus $T(v)$ is non-increasing in v for all $v \in [\underline{v}, \overline{v}]$ and $q(v) > 0$. Now we show that $T(v)$ is either a constant for some interval or $T(v) = vd(v)$ for an interval where $vd(v)$ is non-increasing. Consider the solution to the following relaxed problem:

$$\max_{T(\cdot) \text{ non-increasing}} \int_{T(v) \leq vd(v)} T(v) - cd(v)F(v). \tag{3}$$

For any non-increasing $T(\cdot)$, if there exists an interval $[v_1, v_2]$ such that $T(v) < vd(v)$ for all $v \in [v_1, v_2]$ and $T(V)$ is not a constant, consider the following modification such that $T'(v) = \min\{T(v_1), vd(v)\}$ for all $v \in [v_1, v_2]$. Then $T'(v) > T(V)$ and is still non-increasing. So the solution to (3) satisfies that $T(v)$ is either a constant for some interval or $T(v) = vd(v)$ for an interval where $vd(v)$ is non-increasing.

We now show that solution satisfies IC and hence is the solution to the original problem. First, within each interval where $T(\cdot)$ is a constant, consumers have no incentive to deviate to another type within the interval. Second, within each interval where $T(v) = vd(v)$, consumers have no incentive to deviate to another type within the interval. To see this, for $v' < v$, $v'd(v) - T(v) = v'd(v) - vd(v) < 0$; for $v' > v$, $v' \min\{d(v), d(v')\} - T(v) = v'd(v') - vd(v) < 0$. Across the intervals, for v to deviate to a higher type v', $v \min\{d(v), d(v')\} - T(v') = vd(v') - v'd(v') < 0$; for v to deviate to a lower type v', $v \min\{d(v), d(v')\} - T(v') = vd(v) - T(v') < vd(v) - T(v)$.

A.7 Proof of Proposition 7

When $d(v)$ is decreasing and $vd(v)$ is increasing, by Corollary 5, the optimal mechanism is a fixed fee $\tilde{p}$. There exists $v < v'$ such that $vd(v) > \tilde{p}$ and $v'd(v') > \tilde{p}$. The difference in the net surpluses obtained by a consumer of type v and a consumer of type v' is

$$\begin{aligned} & vd(v) - \tilde{p} - \left(v'd(v') - \tilde{p}\right) \\ =& vd(v) - v'd(v') \\ <& 0 \end{aligned}$$

since $vd(v)$ is increasing in v.

A.8 Proof of Proposition 8

Let $M = (q, T(q))_q$ be any menu that consists of more than one quantity bundle. Let θ^M be type of the consumer that has the lowest type and purchases a bundle in M. We show that the menu with a single bundle $(\overline{q}, u(\theta^M, \overline{q}))$ generates more revenue than M. If $u(\theta^M, \overline{q}) \geq T(\overline{q})$, we are done. If $u(\theta^M, \overline{q}) < T(\overline{q})$, there must exists a bundle $(q', T(q')) \in M$ such that

$$u(\theta^M, \overline{q}) - u(\theta^M, q') < T(\overline{q}) - T(q')$$

Since $u_\theta(\theta, q) \geq 0$ and $u_{q\theta}(\theta, q) \leq 0$, for all $\theta' > \theta$,

$$u(\theta', \overline{q}) - u(\theta', q') < T(\overline{q}) - T(q').$$

So all consumers would purchase a bundle other than $\overline{q}$ in M. Assume there exists a bundle q'' with $T(q'') > u(\theta^M, \overline{q})$. Then $T(q'') - T(q') > u(\theta^M, \overline{q}) - u(\theta^M, q') > u(\theta', q'') - u(\theta', q')$. q'' cannot be the preferred bundle than q' to any type $\theta' > \theta^M$. Therefore any menu consists of more than a single bundle of quantity $\overline{q}$ cannot be optimal. The second part of the statement follows from this.

A.9 Proof of Proposition 9

Proof. We begin by solving the relaxed problem when the seller's problem is to optimally sell each q-th unit. The solution is to have $t(d) = \hat{v}$ for all d. Now we just need to show that $t(d) = \hat{v}$ for all d also satisfies the full IC constraint: for all consumers with type (v, d) and $v > \hat{v}$, they purchase d units at the per unit price $\hat{v}$ and all the other consumers do not purchase anything.

Consider any (v, d) with $v < \hat{v}$. This consumer has no incentive to purchase any $d' > d$. If the consumer purchase $d' < d$, the consumer's payoff is $vd' - \hat{v}d' < 0$. Consider any (v, d) with $v > \hat{v}$. This consumer has no incentive to purchase any $d' > d$. If the consumer purchase $d' < d$, the consumer's payoff is $vd' - \hat{v}d' < vd - \hat{v}d$. □